

Lab – Install a Virtual Machine on a Personal Computer

Objectives

Part 1: Prepare a Computer for Virtualization

Part 2: Import a Virtual Machine into VirtualBox Inventory

Background / Scenario

Computing power and resources have increased tremendously over the last 10 years. A benefit of having multicore processors and large amounts of RAM is the ability to use virtualization. With virtualization, one or more virtual computers can operate inside a single physical computer. Virtual computers that run within physical computers are called virtual machines. Virtual machines are often called guests, and physical computers are often called hosts. Anyone with a modern computer and operating system can run virtual machines.

A virtual machine image file has been created for you to install on your computer. In this lab, you will download and import this image file using a desktop virtualization application, such as VirtualBox.

Required Resources

- Computer with a minimum of 2 GB of RAM and 8 GB of free disk space
- High speed Internet access to download Oracle VirtualBox and the virtual machine image file

Note: The image file is about 2.5 GB, and can grow up to 5 GB after the virtual machine is in operation. While you can delete the image file after the virtual machine is imported, the 8 GB free disk space requirement is for users who decide to keep the image file.

Note: To install and run 64bit virtual machines on a host physical computer, the computer needs to be a 64bit system and have hardware virtualization technology enabled in BIOS. If you are unable to install the virtual machine image you may need to reboot your computer and enter setup mode in BIOS to enable hardware virtualization technology under advanced system settings.

Part 1: Prepare a Host Computer for Virtualization

In Part 1, you will download and install desktop virtualization software, and also download an image file that can be used to complete labs throughout the course. For this lab, the virtual machine is running Linux.

Step 1: Download and install VirtualBox

VMware Workstation Player and Oracle VirtualBox are two virtualization programs that you can download and install to support the image file. In this lab, you will use VirtualBox.

- Navigate to <http://www.oracle.com/technetwork/server-storage/virtualbox/downloads/index.html>.
- Choose and download the appropriate installation file based on your operating system.
- When you have downloaded the VirtualBox installation file, run the installer and accept the default installation settings.

Step 2: Download the Virtual Machine image file

The image file was created in accordance with the Open Virtualization Format (OVF). OVF is an open standard for packaging and distributing virtual appliances. An OVF package has several files placed into one directory. This directory is then distributed as an OVA package. This package contains all of the OVF files necessary for the deployment of the virtual machine. The virtual machine used in this lab was exported in accordance with the OVF standard.

Click [here](#) to download the virtual machine image file.

Note: This file is 2.5 GB in size, and it may take over an hour to download, depending on the speed of your Internet connection.

Part 2: Import the Virtual Machine into the VirtualBox Inventory

In Part 2, you will import the virtual machine image into VirtualBox and start the virtual machine.

Step 1: Import the virtual machine file into VirtualBox

- Open **VirtualBox**. Click **File > Import Appliance...** to import the virtual machine image.
- A new window will appear. Specify the location of the .OVA file.
- The appliance settings appear. Check the **Reinitialize the MAC address of all network cards** box at bottom of the window. Leave all other settings as default. Click **Import**.
- When the import process is complete, you will see the new Virtual Machine added to the VirtualBox inventory in the left panel. The virtual machine is now ready to use.

Step 2: Start the virtual machine and log in

- In the inventory shown on the left, select the virtual machine you wish to use.
- Click the **Start** button. It is the green arrow located at the top portion of the VirtualBox application window. A new window will appear, and the virtual machine boot process will start.

Note: If the virtual machine fails to start, either disable the USB Controller by going into the virtual machine's settings and unchecking the USB controller setting under USB or go to the VirtualBox download webpage and download and install the Oracle VM VirtualBox Extension Pack.

- When the boot process is complete, the virtual machine will ask for a username and password. Use the following credentials to log into the virtual machine:

Username: cisco

Password: password

You will be presented with a desktop environment: there is a launcher bar at the bottom, icons on the desktop, and an application menu at the top.

Note: The window running the virtual machine is a completely different computer than your host. Functions such as copy and paste will not work between the two without special software tools installed. Notice the keyboard and mouse focus. When you click inside the virtual machine window, your mouse and keyboard will operate the guest operating system. Your host operating system will no longer detect keystrokes or mouse movements. Press the right **CTRL** key to return keyboard and mouse focus to the host operating system.

Step 3: Familiarize yourself with the Virtual Machine

Use the Ubuntu_CyberEss virtual machine you just installed to complete the labs that require Ubuntu in this course. Familiarize yourself with the icons in the list below:

The launcher icons are on the left (from top to bottom):

- Search Tool
- File manager application
- Firefox Web Browser
- LibreOffice Writer, LibreOffice Calc, LibreOffice Impress
- Ubuntu Software Center

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- Amazon
 - System Settings
 - Terminal
 - Trash
- a. Open the terminal application. Type the command **ip address** at the prompt to determine the IP address of your virtual machine.
- What are the IP addresses assigned to your virtual machine?
- 192.0.2.1/32, 192.0.2.2/32, 192.0.2.3/32, 192.0.2.4/32, 192.0.2.5/32,
- b. Locate and launch the web browser application. Can you navigate to your favorite search engine?
- Yes
- c. Press the right ctrl key to release the cursor from the virtual machine. Now go to the menu at the top of the virtual machine window and choose **File > Close** to close the virtual machine. What options are available?
- Save the machine state
 - Send the shutdown signal
 - Power off the machine
 - Restore current snapshot 'Snapshot 3' (I am also using virtual machine for another course)
- d. Click the **Save the machine state** radio button and click **OK**. The next time you start the virtual machine, you will be able to resume working in the operating system in its current state.

Reflection

What are the advantages and disadvantages of using a virtual machine?

The advantages that I can see is the GUI and how easy it is to navigate as a first time user. It is straightforward, direct, doesn't have that many external elements to it, and it lets you run other operating systems into your machine. Another thing I found useful is the snapshot feature where you can save and revert back to your save.

The disadvantage is that it is heavy on your computer's RAM, it is also not really a machine that can run graphically heavy programs or games.